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Regime-Conditional Attention Head Selection for Time Series Transformers

Transformer-based models have achieved strong results in long-term time series forecasting [1, 2]. However, these models often use many attention heads, which can increase computational cost and may include redundant parts [6]. Existing attention head pruning methods usually measure the importance of each head with one global score over the whole validation set [7, 8]. This assumes that a head has the same importance for all types of time periods.

This project proposes a regime-aware attention head selection method for time series Transformers. The main hypothesis is that attention head importance may change depending on the temporal regime of the input window. For example, some heads may be more useful during trend-dominant periods, while others may be more useful during seasonality-dominant or noise-dominant periods.

First, validation windows will be analyzed using STL decomposition [5]. Each window will be divided into trend, seasonal, and residual components. Then, each window will be labeled as trend-dominant, seasonality-dominant, or noise-dominant according to the strongest component. After that, the importance of each attention head will be measured separately for each regime. This will be done by masking one head at a time and observing the change in validation loss.

The project will evaluate two pruning policies. In Policy A, a static Temporal Importance Score will be calculated for each attention head by combining its importance across different regimes. Heads with low scores will be pruned, and the model will be fine-tuned. This creates one fixed pruned model.

In Policy B, a dynamic regime-conditional head selection method will be tested. Instead of using the same pruned heads for every input, the model will activate different head subsets depending on the detected regime of the input window[9]. For example, if an input window is trend-dominant, the model will use the heads that are most important for trend-dominant windows. This makes the pruning decision more adaptive to the temporal structure of the data.

The main model will be PatchTST [1], because its attention mechanism works over temporal patches, which fits the idea of temporal regimes. The project will first test the method on ETTh1[3] with a prediction length of 96. If the regime-specific head importance patterns are clear, the experiments will be extended to other prediction lengths and datasets such as ETTm1[3] and Weather[4]. iTransformer[2] may also be used as a generalization test, but it will be discussed carefully because its attention works across variables rather than time steps.

The proposed method will be compared with several baselines: no pruning, random head pruning, global importance pruning, and magnitude-based pruning. The evaluation will use

MSE and MAE for forecasting accuracy, and parameter count, FLOPs, and inference latency for efficiency. The goal is to test whether considering temporal regimes can improve attention head pruning compared with using only one global importance score.

[1] https://github.com/yuqinie98/PatchTST

[2] https://github.com/thuml/iTransformer

[3] https://github.com/zhouhaoyi/ETDataset

[4] https://nixtlaverse.nixtla.io/datasetsforecast/long_horizon2.html

[5] https://www.math.unm.edu/~lil/Stat581/STL.pdf

[6] https://arxiv.org/pdf/1706.03762

[7] https://arxiv.org/pdf/1905.10650

[8] https://arxiv.org/pdf/1905.09418

[9] https://arxiv.org/pdf/2102.04906